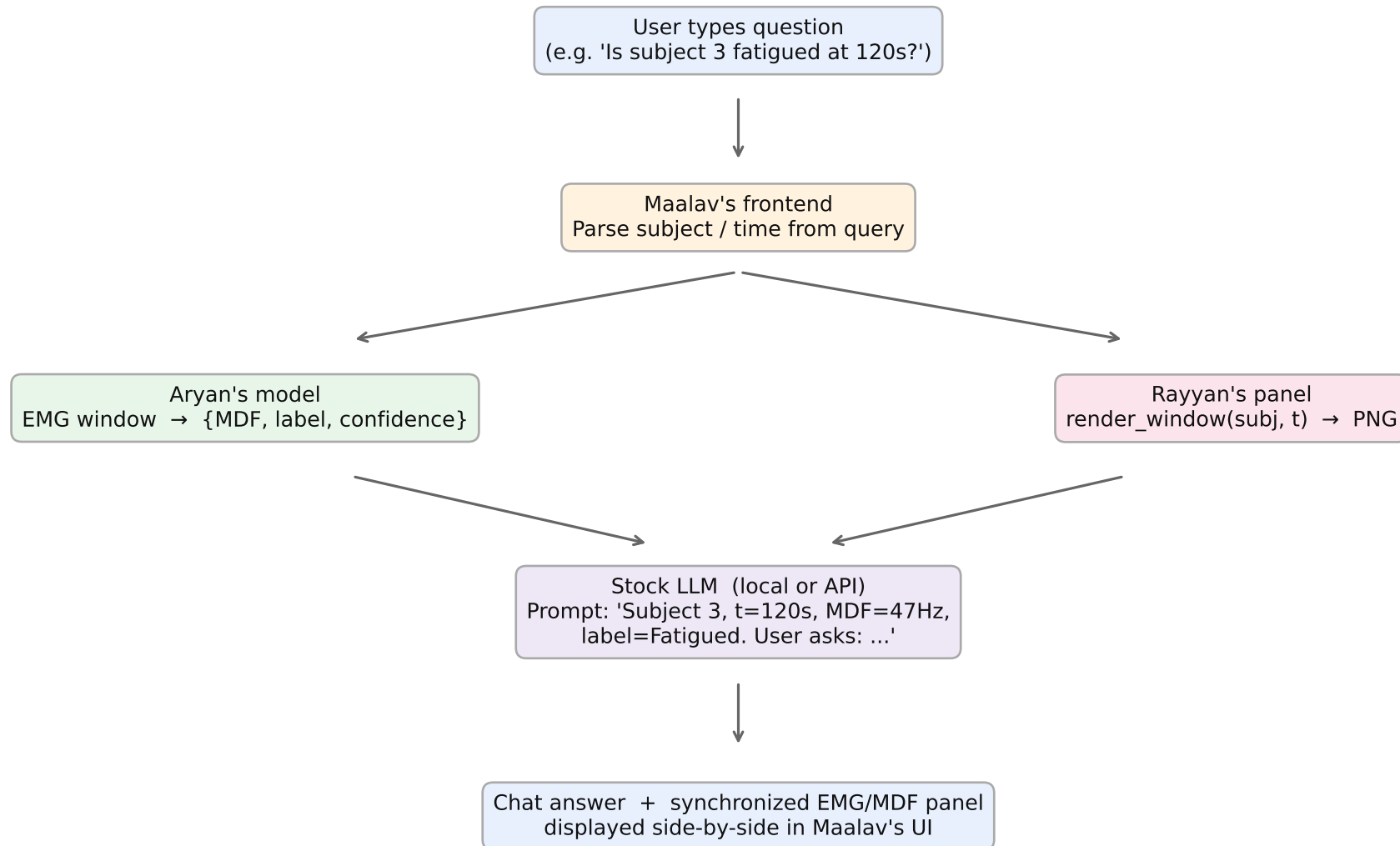


EMG Chatbot System

Project Plan | Signal-Driven AI Systems | 15 June 2026

Architecture



LLM reads structured features (not images). No training required - stock LLM grounded with pipeline outputs.

Per-Person Scope

Rayyan

Own the visual grounding layer of the chatbot.

Expose one callable: `render_window(subject, t_start, side) -> PNG`

Panel shows: EMG signal for the queried window + MDF trend cursor + fatigue label colour.

Output must be embeddable in Maalav's UI (PNG bytes or file path).

The 13-subject overlay viewer is DONE - use for supervisor demos only.

This panel is what makes the chatbot's text answers grounded and visual.

Aryan

Feature extraction layer using Transformers or LSTM.

Input: raw EMG window (subject, t_start, side).

Output: dict {mdf_hz: float, fatigue_label: int, confidence: float}

Wrap as a callable function the frontend can import or call via API.

Existing `classify_biceps.py` pipeline is the foundation - build on top.

Maalav

Frontend chat UI + stock LLM wiring.

Parse user queries for subject / time references.

Call Aryan's `classify()` -> inject features into LLM system prompt.

Call Rayyan's `render_window()` -> display panel alongside LLM text answer.

Stock LLM: Ollama locally (llama3, mistral) or any API - no fine-tuning.

Prompt template: 'Subject N, t=Xs, MDF=YHz, label=Z. User asks: [question]'

Integration Contract (agree this early)

Interface	Signature	Produces	Consumed by
Visual panel	<code>render_window(subj, t_start, side)</code>	PNG bytes / file path	Maalav
Feature extract	<code>classify(subj, t_start) {mdf_hz, fatigue_label, confidence}</code>		Maalav
LLM prompt	<code>build_prompt(features, user_query)</code>	String context for LLM	Maalav
Chat UI	React / HTML frontend	Chat + panel display	End user

Agree signatures before building. A `render_window` that returns a file path and a caller that expects bytes will break silently.

Key Decisions Made

LLM reads structured features, not images

- Multimodal LLMs (llava) cannot reliably extract quantitative biomarkers from waveform images.
- The pipeline already produces exact numbers - MDF, fatigue label, confidence.
- Feed those as text context. The graph stays as a human-facing visual only.
- Decision: structured features -> LLM prompt.

No model training required

- 'Use our own models' means the LSTM/Transformer fatigue classifier is the team's model.
- The LLM is a stock reasoning layer on top (Ollama, llama3, mistral, or any API).
- No fine-tuning. No training data collection. Plug the stock LLM in and prompt it.
- Decision: stock LLM + structured context injection.

Rayyan's task: visual grounding layer, not standalone viewer

- The 13-subject overlay is done and useful for demos. It is not the chatbot's core.
- The chatbot needs a per-query panel: show the specific subject/window being discussed.
- This makes the LLM's text answer grounded - user sees exactly what the model describes.
- Decision: `render_window(subj, t, side)` -> PNG, embedded in Maalav's UI.